P09 Project Plan

Visual Similarity Search using Self-Supervised Embeddings

Stage 1 — due 6 October 2026

1. Project scope

How do the retrieval quality and speed of pretrained DINO and DINOv2 embeddings compare with supervised embeddings on Oxford-IIIT Pet?

The team will build a query-by-image system: given an image, it will return the five most similar images from a fixed gallery. The task covers all 37 pet breeds. We will compare four pretrained feature extractors: DINO ViT-S/16, supervised DeiT-S/16, DINOv2 ViT-B/14, and supervised DeiT-B/16. All models will use the same gallery, queries, relevance rule, and evaluation code. Differences in architecture and pretraining data will be considered when interpreting the results.

Pretrained model sources: DINO, DINOv2, and DeiT.

2. Dataset preparation

Oxford-IIIT Pet contains images of cats and dogs from 37 breeds, with roughly 200 images per breed. We will use the full dataset, with no breed subset. Its breed labels provide a measurable proxy for visual similarity; images from different breeds may still look alike. Data and labels will be loaded with `torchvision.datasets.OxfordIIITPet` using `target_types="category"`.

Sources: Oxford dataset description and files; Torchvision loader and official splits.

The official `trainval` split will be divided equally by breed. The official `test` split will remain untouched.

Split parameters: `test_size=0.5`, `random_state=42`, and `stratify=labels`. Images will be sorted by ID before splitting, and the resulting `image_id`, `breed`, and `split` manifest will be saved for the whole team. We plan to use scikit-learn's train_test_split.

Project split	Source	Use
Train / gallery	First half of `trainval`	Images indexed for retrieval; their embeddings are computed in advance.
Validation queries	Second half of `trainval`	Check the pipeline and choose settings before final evaluation.
Test queries	Official `test`	Final evaluation and examples for the demo.

No query image will appear in the gallery. The frozen models will not be trained on these splits: "train" denotes the retrieval gallery. Preprocessing will be deterministic and identical for gallery and query images: RGB conversion, resize the shorter side to 256 pixels, center crop to 224×224, convert to a tensor, and normalize with ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). No random augmentation will be used.

3. Experimental plan

All four pretrained backbones will remain frozen; there is no fine-tuning. For the supervised DeiT models, features will be taken before the classification head rather than from class logits. Each image embedding will be L2-normalized, and gallery images will be ranked by cosine similarity to the query embedding. Validation queries will be used for model and setting choices; the test queries will be used for final reporting.

An image is relevant to a query when both have the same breed label. $P@K$ is the fraction of relevant images among the first K results; $R@K$ is the fraction of all relevant gallery images found within the first K results. Scores will be averaged over queries. Labels are used for evaluation, not for retrieval.

Measure	Purpose
Precision@5 and Precision@10	Fraction of same-breed images among the top 5 and top 10 results.
Recall@5 and Recall@10	Fraction of all relevant gallery images retrieved in the top 5 and top 10.
Hit@5	Fraction of queries with at least one same-breed image in the top 5.
Gallery build time, s	Total time to extract and store all gallery embeddings.
Query processing time, ms	Median embedding extraction time and top-5 search time, reported separately.

Runtime will be measured on the same device and query set for all four models. Final results will include comparison tables or figures and at least three unsuccessful retrieval examples with brief analysis.

4. Implementation plan

1. Load images and labels, create the fixed split manifest, and implement deterministic preprocessing.
2. Load pretrained weights, remove supervised classification heads, extract and store normalized gallery embeddings for each model.
3. Implement a shared `query(image, model, k)` function using cosine similarity and top-K ranking.
4. Compute quality and runtime measures and save comparison tables and failure examples.
5. Build a small Gradio interface for image upload, model selection, and a top-5 image gallery.

The main tools are PyTorch, Torchvision, scikit-learn, NumPy, `timm`, and Gradio; pretrained DINO/DINOv2/DeiT weights are available from the linked repositories.

5. Team responsibilities

Stage	Deadline	Owner	Deliverable
1. Project plan	6 Oct	Danil	Document the question, dataset choice, split design, metric definitions, runtime measurement plan, team tasks, and demo scenario.
2. First models	13 Oct	Azamat	Load and split the dataset, build the gallery, run DINO ViT-S/16 and supervised DeiT-S/16, implement top-K retrieval, and report initial metrics and times.
3. Second models	20 Oct	Karim	Add DINOv2 ViT-B/14 and supervised DeiT-B/16 to the same retrieval interface; report metrics and times on the same split and gallery.
4. Progress check	27 Oct	Aslan	Prepare a preliminary comparison of all four models and list the remaining experiments.
5. Results checkpoint	3 Nov	Aslan	Collect final quality and runtime results, tables or figures, and at least three failed retrieval examples with explanations.
6. Final submission	10 Nov	Danil	Assemble the two-page technical summary, reproducible code, README, environment details, slides, demo, and member contributions.

Demo: upload an image, choose one of the four models, and display the five nearest gallery images with similarity scores. Gallery embeddings will be precomputed; the query embedding will be computed when the image is uploaded.